

A NEW CLASS OF COPULAS WITH TAIL DEPENDENCE

*Matthias Fischer*¹

University of Erlangen-Nuremberg, Department of Statistics & Econometrics

e-mail: *Matthias.Fischer@fau.de*

and

Gerd Hinzmann

University of Erlangen-Nuremberg

Key words: Geometric mean; arithmetic mean; tail dependence

Summary: Copula-based multivariate models allow to specify the marginal distributions separately from the dependence structure (i.e. the copula) which links these distributions to form a joint distribution. Within this work we introduce a new family of copulas ("generalized mean copulas") which is positive comprehensive (i.e. includes both independence and maximum dependence) and allows for upper tail dependence. It includes the Spearman copula and a specific Fréchet copula as special cases. Some properties of the new copula are derived.

1. Introduction

Since the pioneering work of Embrechts, McNeil and Straumann (1999) and the research group of Credit Lyonnais (e.g. Bouyé et al., 2000), the popularity of the copula concept in finance steadily increases. As it was demonstrated by Sklar (1959), each multivariate probability distribution can be decomposed into its margins and its dependence structure, e.g. by its copula. In contrast to the modelling of the marginal distribution, capturing the adequate dependence structure between the assets under consideration is still a challenging task. Above that, risk managers are especially faced with the problem that assets tend to collapse together, a phenomenon titled as "tail dependence" in the literature. This gives rise to two issues. Firstly, the construction of copulas which are able to assign sufficient probability to these extreme events of common rises or falls. Secondly, the development of simple and accurate tail dependence estimators (TDE) in order to check whether tail dependence is present in a data set. Based on a specific copula which is introduced within this work, we are able to give some contribution to each of these aspects.

The structure of the paper is as follows. Section 2 briefly reviews the copula concept and the notion of tail dependence. In Section 3, the generalized mean copulas are introduced and some properties are derived. The main focus of Section 4 is on a special case, the so-called harmonic mean copulas. Finally, Section 5 is dedicated to the derivation of the tail dependence coefficient.

¹ Corresponding author.

2. Copulas: A review

Let $[a, b] \subseteq \mathbb{R}$. A function $K : [a, b] \times [a, b] \rightarrow \mathbb{R}$ is said to be *2-increasing* if its K -volume

$$V_K(u_1, u_2, v_1, v_2) \equiv K(u_2, v_2) - K(u_2, v_1) - K(u_1, v_2) + K(u_1, v_1) \geq 0 \quad (1)$$

for all $a \leq u_1 \leq u_2 \leq b$ and $a \leq v_1 \leq v_2 \leq b$. If, additionally, $[a, b] = [0, 1]$ and K satisfies the boundary conditions

$$K(u, 0) = K(0, v) = 0, \quad K(u, 1) = u \quad \text{and} \quad K(1, v) = v \quad (2)$$

for arbitrary $u, v \in [0, 1]$, K is commonly termed a copula and we write C , instead.

Putting it in a different way, let X and Y denote two random variables with joint distribution $F_{X,Y}(x, y)$ and continuous marginal distribution functions $F_X(x)$ and $F_Y(y)$. According to Sklar's (1959) fundamental theorem, there exists a unique representation

$$F_{X,Y}(x, y) = C(F_X(x), F_Y(y))$$

of the joint distribution into its marginal distribution functions and the so-called copula

$$C(u, v) = P(U \leq u, V \leq v), \quad U \equiv F_X(X), \quad V \equiv F_Y(Y)$$

on $[0, 1]^2$ which comprises the information about the underlying dependence structure. For details on copulas we refer to Nelsen (2006) and Joe (1999).

Prominent examples are the *independence* copula

$$C_I(u, v) = uv \quad (3)$$

which corresponds to bivariate distributions with independent marginals and the maximum copula

$$C_U(u, v) = \min\{u, v\}, \quad (4)$$

associated to random variables which are co-monotone and, thus, constituting an upper bound for all copulas. Copulas which include both the independent copula (i.e. no dependence) and the maximum copula (i.e. perfect dependence) will be called *positive comprehensive* (see, e.g. Joe, 1999), henceforth. Examples are given in the next section.

3. A generalized mean copula

In general, every convex-combination of two (or more) copulas is again a copula. For instance, the convex combination of (3) and (4) corresponds to family B11 in Joe (1999), i.e.

$$C_A(u, v; \alpha) = \alpha \min\{u, v\} + (1 - \alpha)uv, \quad \alpha \in [0, 1] \quad (5)$$

which is also a special case of the Fréchet family (Fréchet, 1958). In other words, C_A is a (weighted) arithmetic mean of C_I and C_U and henceforth referred to as the arithmetic mean copula. Similarly,

family B12 in Joe (1999) – also known as the Spearman or Cuadras-Augé copula (Cuadras and Augé, 1981) – is given by

$$C_G(u, v; \alpha) = \min\{u, v\}^\alpha (uv)^{1-\alpha}, \quad \alpha \in [0, 1] \quad (6)$$

which can be seen as a weighted geometric mean of C_I and C_U . In both cases $\alpha = 0$ results in the independence copula, whereas $\alpha = 1$ results in the maximum copula. Hence, both copula families are positive comprehensive. A natural generalization of (5) and (6) is given by a weighted Hölder (or power) mean of C_I and C_U . This generalized mean, also known as Hölder mean, is an abstraction of the (weighted) arithmetic and (weighted) geometric means (see, e.g. Borwein and Borwein, 1987; Bullen, 2003). For $m \in \mathbb{R}$ and $\alpha \in [0, 1]$, consider

$$K(u, v; \alpha, m) \equiv (\alpha \min\{u, v\}^m + (1 - \alpha)(uv)^m)^{1/m}. \quad (7)$$

Note that $K(u, v; \alpha, 1) = C_A(u, v)$ and – after taking the limit – $K(u, v; \alpha, 0) = C_G(u, v)$. We next show that K is a copula for $m \in \mathbb{R} \setminus \{0, 1\}$, too. The boundary conditions are easily verified: $K(u, 0; \alpha, m) = K(0, v; \alpha, m) = 0$, $K(u, 1; \alpha, m) = u$ and $K(1, u; \alpha, m) = v$ for all $u, v \in [0, 1]$. Moreover, K is an exchangeable function, i.e. $K(u, v; \alpha, m) = K(v, u; \alpha, m)$. In order to prove that K is actually a copula it remains to verify that K satisfies the two-increasing condition from (1) with

$$\begin{aligned} V_K = & (\alpha \min\{u_2, v_2\}^m + (1 - \alpha)(u_2 v_2)^m)^{\frac{1}{m}} - (\alpha \min\{u_2, v_1\}^m + (1 - \alpha)(u_2 v_1)^m)^{\frac{1}{m}} \\ & - (\alpha \min\{u_1, v_2\}^m + (1 - \alpha)(u_1 v_2)^m)^{\frac{1}{m}} + (\alpha \min\{u_1, v_1\}^m + (1 - \alpha)(u_1 v_1)^m)^{\frac{1}{m}}. \end{aligned}$$

Lemma 1 $K(u, v; \alpha, m)$ from (7) is a copula for $\alpha \in [0, 1]$ and $m \in \mathbb{R}$.

Proof. In order to prove that $V_K \geq 0$, consider the following cases

- **Case 1:** $u_1 \leq v_1 \leq u_2 \leq v_2$,
- **Case 2:** $u_1 \leq v_1 \leq v_2 \leq u_2$,
- **Case 3:** $u_1 \leq u_2 \leq v_1 \leq v_2$,
- **Case 4:** $v_1 \leq u_1 \leq v_2 \leq u_2$,
- **Case 5:** $v_1 \leq u_1 \leq u_2 \leq v_2$,
- **Case 6:** $v_1 \leq v_2 \leq u_1 \leq u_2$.

Introducing the auxiliary function

$$f(u) \equiv f(u; \alpha, m) \equiv (\alpha + (1 - \alpha)u^m)^{1/m}$$

we have to check the validity of the following inequalities:

- **Case 1:** $u_2 f(v_2) - v_1 f(u_2) - u_1 f(v_2) + u_1 f(v_1) \geq 0$,
- **Case 2:** $v_2 f(u_2) - v_1 f(u_2) - u_1 f(v_2) + u_1 f(v_1) = (v_2 - v_1)f(u_2) - u_1(f(v_2) - f(v_1)) \geq 0$,
- **Case 3:** $u_2 f(v_2) - u_2 f(v_1) - u_1 f(v_2) + u_1 f(v_1) = (u_2 - u_1)(f(v_2) - f(v_1)) \geq 0$,

- **Case 4:** $v_2 f(u_2) - v_1 f(u_2) - u_1 f(v_2) + v_1 f(u_1) \geq 0$,
- **Case 5:** $u_2 f(v_2) - v_1 f(u_2) - u_1 f(v_2) + v_1 f(u_1) = (u_2 - u_1) f(v_2) - v_1 (f(u_2) - f(u_1)) \geq 0$,
- **Case 6:** $v_2 f(u_2) - v_1 f(u_2) - v_2 f(u_1) + v_1 f(u_1) = (v_2 - v_1) (f(u_2) - f(u_1)) \geq 0$.

Actually, using the exchangeability of K , it suffices to prove case 1, case 2 and case 3.

First notice that for $\Delta \geq 0$ it holds that $(*)$ $uf(u + \Delta) \leq (u + \Delta)f(u)$: For $m > 0$, we observe the equivalence $uf(u + \Delta) \leq (u + \Delta)f(u) \iff$

$$\begin{aligned} u(\alpha + (1 - \alpha)(u + \Delta)^m)^{1/m} &\leq (u + \Delta)(\alpha + (1 - \alpha)u^m)^{1/m} \iff \\ u^m(\alpha + (1 - \alpha)(u + \Delta)^m) &\leq (u + \Delta)^m(\alpha + (1 - \alpha)u^m) \iff u^m \leq (u + \Delta)^m. \end{aligned}$$

The derivation for $m < 0$ is similar, but now $u^m \geq (u + \Delta)^m$.

Case 1: Assume that $u_1 \leq v_1 \leq u_2 \leq v_2$. From $(*)$, $v_1 f(u_2) \leq u_2 f(v_1)$ and it follows that

$$u_2 f(v_2) - v_1 f(u_2) - u_1 f(v_2) + u_1 f(v_1) \geq u_2 f(v_2) - u_2 f(v_1) - u_1 f(v_2) + u_1 f(v_1)$$

and $u_2 f(v_2) - u_2 f(v_1) - u_1 f(v_2) + u_1 f(v_1) = (u_2 - u_1)(f(v_2) - f(v_1)) \geq 0$.

Case 2: Assuming that $u_1 \leq v_1 \leq v_2 \leq u_2$ we have to show that

$$(v_2 - v_1)f(u_2) - u_1(f(v_2) - f(v_1)) \geq 0.$$

We restrict ourselves to $0 < v_1 < v_2$ because $v_1 = v_2$ and $u_1 = v_1 = 0$ is trivial. Now $f(u_2) \geq f(v_2)$ (because f is strictly monotone increasing for $u \in [0, 1]$) and $u_1 \leq v_1$ and therefore

$$(v_2 - v_1)f(u_2) - u_1(f(v_2) - f(v_1)) \geq (v_2 - v_1)f(v_2) - v_1(f(v_2) - f(v_1))$$

and it suffices to show that

$$(v_2 - v_1)f(v_2) - v_1(f(v_2) - f(v_1)) \geq 0 \iff \frac{f(v_2)}{v_1} \geq \frac{f(v_2) - f(v_1)}{v_2 - v_1}.$$

Denoting $v_2 = v_1 + \Delta$ and $u = v_1$, we can rewrite the last inequality to

$$\frac{f(v_1 + \Delta)}{v_1} \geq \frac{f(v_1 + \Delta) - f(v_1)}{\Delta} \iff uf(u + \Delta) \leq \Delta f(u + \Delta) + uf(u)$$

which can be derived again from $(*)$ and because f is monotone increasing. ■

Contour lines of generalized mean copulas are given in Figure 1.

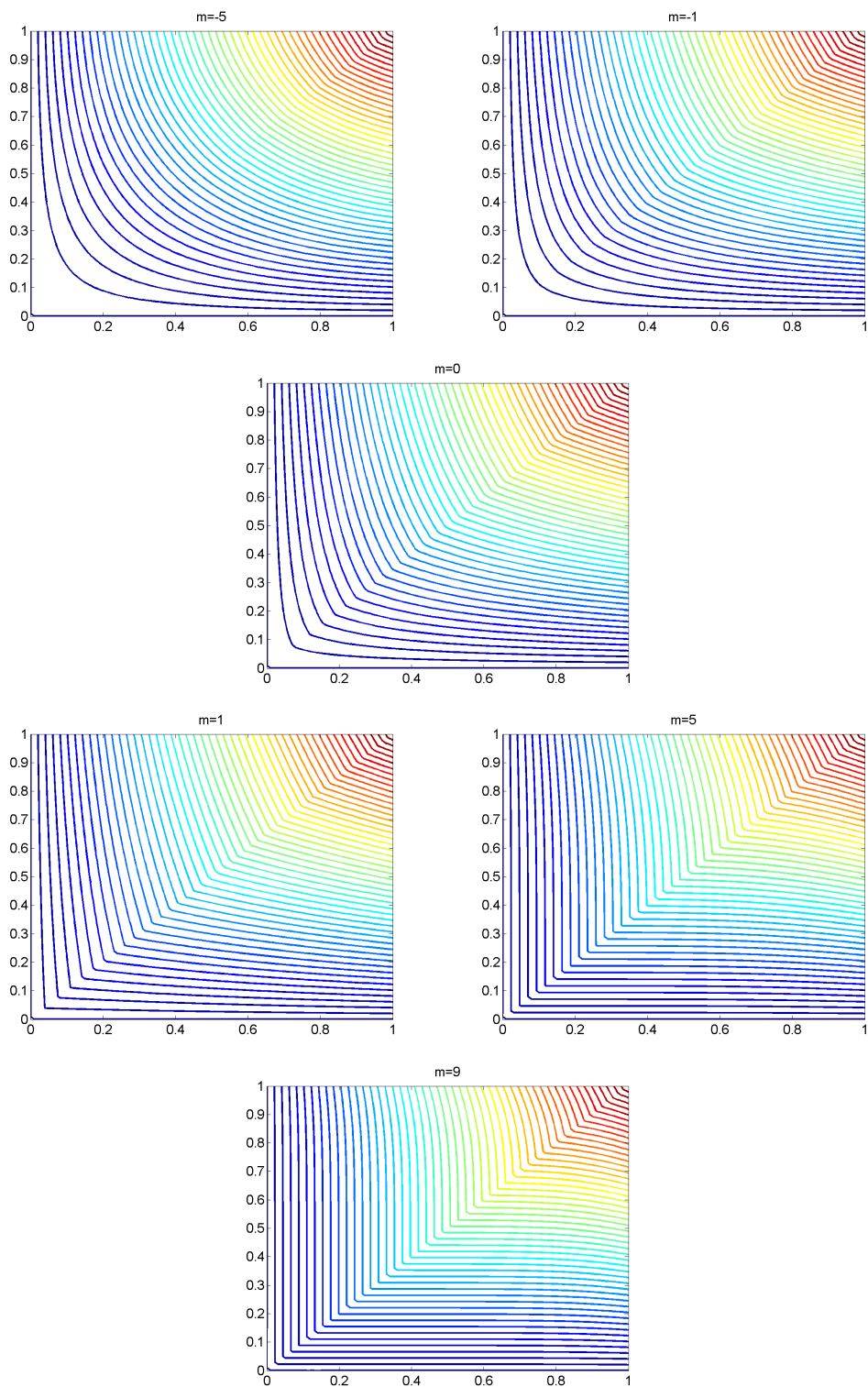


Figure 1: Contour lines of generalized mean copulas for different values of m and $\alpha = 0.5$.

4. A special case: The harmonic mean copula

We now restrict ourselves to a specific member of the generalized mean copula family, namely the so-called *harmonic mean copulas* (HMC) which emerge for $m = -1$, i.e.

$$C_H(u, v; \alpha) \equiv \frac{1}{\frac{\alpha}{\min\{uv\}} + \frac{1-\alpha}{uv}} = \frac{\min\{u, v\} \cdot uv}{\alpha uv + (1-\alpha) \min\{u, v\}} = \frac{C_G(u, v; 0.5)^2}{C_A(u, v; \alpha)}. \quad (8)$$

For the HMC-copulas global dependence measures like Kendall's τ , Spearman's ρ , Gini's γ and Blomqvist's β can be derived explicitly. For a detailed treatment of dependence measure we refer to Mari and Kotz (2004).

Lemma 2 For the harmonic mean copula $C_H(u, v; \alpha)$ we obtain

$$\begin{aligned} \tau &= \tau(\alpha) = \frac{18\alpha^2 - 12\alpha - 4\alpha^3 - \alpha^4 + (24\alpha - 12\alpha^2 - 12) \ln(1-\alpha)}{\alpha^4} \\ \rho_S &= \rho_S(\alpha) = \frac{12\alpha - 30\alpha^2 + 22\alpha^3 - 3\alpha^4 + (12 - 36\alpha + 36\alpha^2 - 12\alpha^3) \ln(1-\alpha)}{\alpha^4}, \\ \gamma &= \gamma(\alpha) = \frac{3\alpha^2 - 2\alpha^3 + \ln(1-\alpha/2)(8-8\alpha) - \ln(1-\alpha)(4-8\alpha+4\alpha^2)}{\alpha^3}, \\ \beta &= \beta(\alpha) = \frac{\alpha}{2-\alpha}. \end{aligned}$$

Proof. The main obstacle is to get rid of the minimum expression in the definition of C_H in (8). Following Cherubini, Luciano and Vecchiato (2004), Chapter 3.1, Kendall's τ for copulas which are singular or have both an absolutely continuous and a singular component can be computed by

$$\begin{aligned} \tau &= 1 - 4 \int_0^1 \int_0^1 \frac{\partial C(u, v)}{\partial u} \frac{\partial C(u, v)}{\partial v} du dv \\ &= 1 - 4 \int_0^1 \left[\int_0^v \frac{\partial \left(\frac{uv}{\alpha v + 1 - \alpha} \right)}{\partial u} \frac{\partial \left(\frac{uv}{\alpha v + 1 - \alpha} \right)}{\partial v} du + \int_v^1 \frac{\partial \left(\frac{uv}{\alpha u + 1 - \alpha} \right)}{\partial u} \frac{\partial \left(\frac{uv}{\alpha u + 1 - \alpha} \right)}{\partial v} du \right] dv \\ &= 1 - 4 \int_0^1 \left[\int_0^v -\frac{vu(-1+\alpha)}{(\alpha v + 1 - \alpha)^3} du - \int_v^1 \frac{vu(-1+\alpha)}{(\alpha u + 1 - \alpha)^3} du \right] dv \\ &= 1 - 4 \int_0^1 \frac{v(-1+\alpha)(v^3\alpha - 3\alpha v - 1 + 2\alpha + \alpha^2 v^3 - 3\alpha^2 v^2 + 3v\alpha^2 - \alpha^2)}{(\alpha v + 1 - \alpha)^3} dv \\ &= \frac{18\alpha^2 - 12\alpha - 4\alpha^3 - \alpha^4 + (24\alpha - 12\alpha^2 - 12) \ln(1-\alpha)}{\alpha^4}. \end{aligned}$$

The result on Spearman's ρ follows with

$$\begin{aligned} \rho_S &= 12 \int_0^1 \int_0^1 C_H(u, v; \alpha) du dv - 3 \\ &= 12 \int_0^1 \left(\int_0^v \frac{uv}{\alpha v + 1 - \alpha} du + \int_v^1 \frac{uv}{\alpha u + 1 - \alpha} du \right) dv - 3 \end{aligned}$$

and some tedious but straightforward calculations. Similarly, Gini's γ follows from

$$\begin{aligned}\gamma &= 4 \int_0^1 C(u, 1-u) + C(u, u) du - 2 \\ &= 4 \left(\int_0^{0.5} \frac{u(1-u)}{1-\alpha u} du + \int_{0.5}^1 \frac{u(1-u)}{\alpha u + (1-\alpha)} du + \int_0^1 \frac{u^2}{\alpha u + (1-\alpha)} du \right) - 2.\end{aligned}$$

Finally, Blomquist's β is simply $\beta = 4C_H(0.5, 0.5; \alpha) - 1$. ■

Figure 2 illustrates the dependence of τ, ρ and γ on the parameter α . All curves are strictly monotone increasing and convex. The results from the last lemma may be useful to estimate the parameters of the harmonic mean copulas.

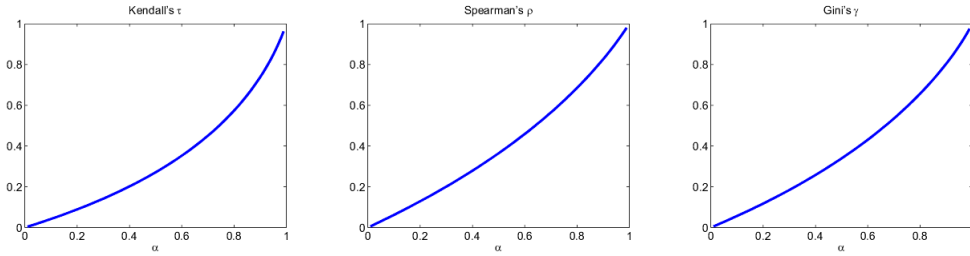


Figure 2: Influence of α on different dependence measures.

5. Derivation of the (upper) tail dependence coefficient

The concept of tail dependence provides, roughly speaking, a measure for extreme co-movements in the lower and upper tail of $F_{X,Y}(x,y)$, respectively and is very useful in financial risk management. Regarding the generalized mean copulas, we focus on the upper tail dependence coefficient (TDC) which is usually defined by

$$\lambda_U \equiv \lim_{u \rightarrow 1^-} P(Y > F_Y^{-1}(u) | X > F_X^{-1}(u)) = \lim_{u \rightarrow 1^-} \frac{1 - 2u + C(u, u)}{1 - u} \in [0, 1], \quad (9)$$

noting that λ_U is solely depending on the copula and not on the marginal distributions. Coles, Heffernan and Tawn (1999) provide an equivalent version of (9),

$$\lambda_U = 2 - \lim_{u \rightarrow 1^-} \frac{\log C(u, u)}{\log(u)}. \quad (10)$$

Fischer and Dörfinger (2010) showed that $C_G(u, v)$ from (6) is upper tail dependent with TDC $\lambda_U = \alpha \in [0, 1]$. This result also holds for the generalized mean copula, as the next lemma shows.

Lemma 3 The (upper) TDC of the generalized mean copula is given by $\lambda_U = \alpha$.

Proof. Plugging (7) into (10) and applying l'Hospital's rule, we obtain

$$\begin{aligned}\lambda_U &= 2 - \frac{1}{m} \lim_{u \rightarrow 1^-} \frac{\log(\alpha u^m + (1 - \alpha)u^{2m})}{\log(u)} \\ &= 2 - \frac{1}{m} \lim_{u \rightarrow 1^-} \frac{u(m\alpha u^{m-1} + 2m(1 - \alpha)u^{2m-1})}{\alpha u^m + (1 - \alpha)u^{2m}} = 2 - (2 - \alpha) = \alpha.\end{aligned}$$

■

Hence, the upper TDC λ_U is solely determined by the parameter α and not by m .

Acknowledgement

The authors thank an anonymous referee for his/her helpful comments and suggestions which significantly improved the presentation of the paper.

References

- BORWEIN, J. AND BORWEIN, P. (1987). *Pi and the AGM: A Study in Analytic Number Theory and Computational Complexity*. Wiley, New York.
- BOUYÉ, E., DURRLEMAN, V., NIKEGBALI, A., RIBOULET, G., AND RONCALLI, T. (2000). Copulas for finance - a reading guide and some applications. Working paper, Credit Lyonnais.
- BULLEN, P. (2003). *Handbook of Means and their Inequalities*. Kluwer Academic Publishers, Dordrecht.
- CHERUBINI, U., LUCIANO, E., AND VECCHIATO, W. (2004). *Copula Methods in Finance*. John Wiley & Sons, Chichester.
- COLES, S., HEFFERNAN, J., AND TAWN, J. (1999). Dependence measures for extreme value analyses. *Extremes*, **2**, 339–365.
- CUADRAS, C. AND AUGÉ, J. (1981). A continuous general multivariate distribution and its properties. *Communications in Statistics – Theory and Methods*, **10** (4), 339–353.
- EMBRECHTS, P., MCNEIL, A., AND STRAUMANN, D. (1999). Correlation: Pitfalls and alternatives. *RISK Magazine*, **5**, 69–71.
- FISCHER, M. AND DÖRFLINGER, M. (2010). A note on a non-parametric tail dependence estimator. *Far East Journal of Theoretical Statistics*, **32** (1), 1–5.
- FRÉCHET, M. (1958). Remarque au sujet de la note précédente. *Royal Academy of Science, Paris*, **246**, 2719–2720.
- JOE, H. (1999). *Multivariate Models and Dependence Concepts*. Chapman & Hall, London.
- MARI, D. D. AND KOTZ, S. (2004). *Correlation and Dependence*. World Scientific.
- NELSEN, R. (2006). *An Introduction to Copulas*. Springer, Berlin.
- SKLAR, A. (1959). Fonctions de répartition à n dimensions et leurs marges. *Publications de l'Institut de Statistique de l'Université de Paris*, **8** (1), 229–231.